Lorem Ipsum Dolor Sit Amet

Lorem Ipsum Dolor Sit Amet Consectetur
Lorem Ipsum University

Abstract

The task of motion planning is equivariant to rigid-body transformations, but the field the flow matching network learns is not. In the current literature, there are three techniques to restore equivariance: in the data, in the operator, or in the weights.

A motion planning task names a start and a goal, and those two points already determine a canonical frame. No existing work measures what that frame is worth before any mechanism is applied, and none separates how equivariant a trained model became from how much that equivariance bought it.

We ask both. Holding architecture, data and training budget fixed, we build all three mechanisms on one cluttered 3D benchmark and calibrate against a trivial floor and four classical planners on identical held-out problems. The frame removes five of the six degrees of freedom of SE(3) exactly, at initialisation and at no cost, and is worth +36.5 points. The three mechanisms for the sixth are worth under a point each. Two models that are equally equivariant differ by 28 points. We also bound our own result from above: giving each waypoint local geometry is worth more than the frame is, and the frame still adds +24.8 alongside it.

1 Introduction

Motion planning is a fundamental problem in robotics, requiring the generation of collision-free, kinematically feasible trajectories from a start to a goal configuration in complex environments. Equivariance to rigid-body transformations is a cornerstone of generalization in this setting. It ensures that a planner solves structurally identical tasks identically, regardless of where in the world they are posed. Sampling-based [14] and optimization-based [30, 24] algorithms are equivariant by construction, since they act on the geometry directly, and they offer completeness or convergence guarantees. Their per-query computational cost limits real-time applicability. Learning-based planners [22, 8, 12, 3] amortize that cost across queries and have emerged as a promising alternative. They do not inherit the symmetry. A network trained on world coordinates relearns the same manoeuvre at every position and orientation.

Existing work restores it in one of three parts of the pipeline. In the data, through augmentation. In the inference operator, through frame averaging or canonicalization [20, 13]. Or in the weights, through architectures equivariant by construction [5, 26, 9, 6], as adopted in recent robot learning systems [25, 23, 28, 29]. All three modify the model. None measures how much of the symmetry the problem statement already supplies before any model is built.

A planning query names a start and a goal, and those two points determine a frame. Place the origin at their midpoint, align the first axis with the direction between them, and express every waypoint and every obstacle in that frame. Three translational and two rotational degrees of freedom vanish exactly, at initialisation, at no computational cost and with no constraint on the architecture. The six numbers that conditioned the network on the query collapse to one invariant scalar. The frame needs a second axis to be determined, and ours takes it from a fixed world axis, so rotating the whole problem rolls the frame rather than leaving it unchanged. What survives is exactly that roll: SE(3) collapses to SO(2), not to nothing. No continuous rule can fix the remainder [17], so the sixth degree of freedom must be handled by one of the three mechanisms above. Fig. 1 takes the group apart one stage at a time.

We build all three on a cluttered 3D benchmark, hold architecture, data and training budget fixed, and calibrate against a trivial floor and four classical planners on the identical held-out problems. The frame is worth +36.5 points of held-out collision-free rate. The three mechanisms for the sixth are worth +0.8, +0.68 and +0.15. A straight line from start to goal scores 15.6%, and the world-frame model, trained to a plateau, reaches 14.60%.

Our contributions are as follows.

- **Five of six degrees of freedom are free.** A frame derived from the query in closed form removes them exactly, raising the held-out collision-free rate from $14.60 \pm 0.20\%$ to $51.10 \pm 0.25\%$ across three seeds, with no change to the architecture, the data or the budget.
- **All three mechanisms for the sixth, measured against each other.** None recovers a point. We explain that rather than inferring it from flat curves, by measuring the non-equivariance residual of the learned field and proving it bounds what any post-hoc symmetrisation can remove. We then show the residual does not predict what a mechanism is worth: two models that are equally equivariant differ by 28 points.
- **Calibration in both directions, including against ourselves.** Supplying each waypoint with local geometry is worth +40.0 points where the

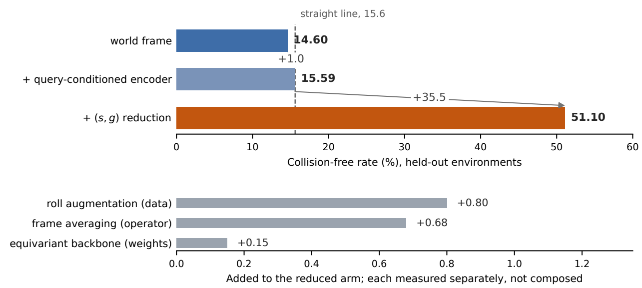

Figure 1: **Sixty environments, trained to convergence, three seeds**, the setting used throughout this paper. *Top*: a frame read off the query in closed form is worth +35.5 points, and showing the obstacle encoder the same query without changing frame is worth +1.0. The dashed line is a straight segment from start to goal on the identical problems, which the world-frame model does not beat. *Bottom*: the three mechanisms for the one degree of freedom the frame cannot remove. Each is measured against its own baseline and they are alternatives rather than a decomposition, so they are not composed here and were never composed in the experiments.

> frame is worth +36.5, so the representation change studied here is not the largest effect available on this benchmark. It is not a substitute for that information either: with local geometry present, the frame still adds +24.8.

2 Related Work

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3 Method

3.1 Decomposing the group

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Proposition 1 (Lorem ipsum). *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua* $X' = R_{\hat{x}}(\theta)\, X$.

Proposition 2 (Dolor sit amet). *Lorem ipsum dolor sit amet consectetur adipiscing elit sed do eiusmod tempor incididunt ut labore.*

Proof. Lorem ipsum dolor sit amet consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua ut enim ad minim veniam. $\square$

4 Setup and Calibration

4.1 Benchmark and protocol

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adipiscing elit	00.0	0.000	0.000
sed do eiusmod	00.0	0.000	0.000
tempor incididunt	00.0	−0.000	0.000
ut labore et dolore	**00.0**	−0.000	0.000

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5 Results

5.1 Scaling

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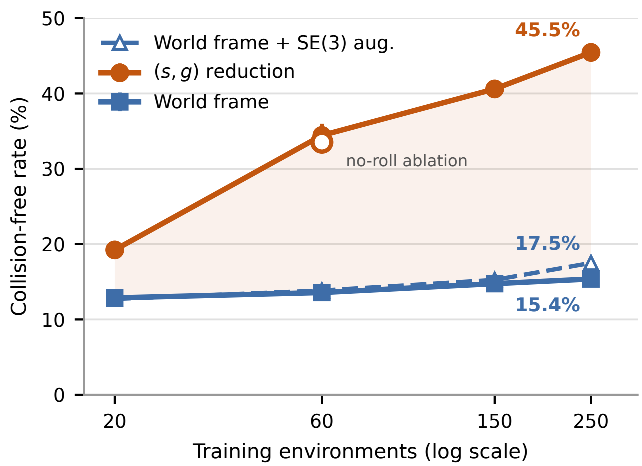

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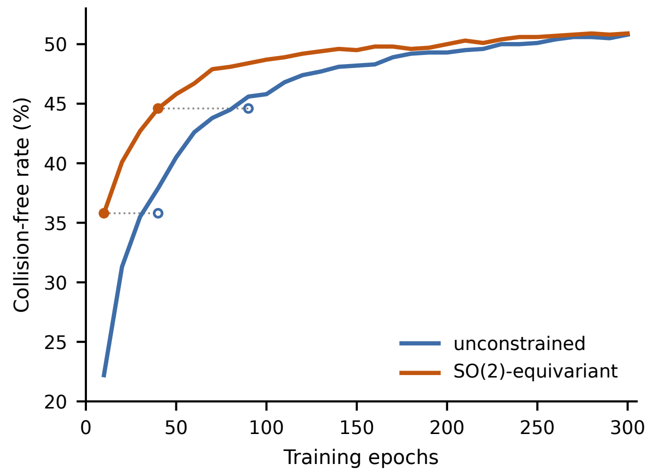

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5.2 Control: a query-conditioned encoder

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5.3 Control: local geometry, and how it scales

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5.4 Replications

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We ask both. Holding architecture, data and training budget fixed, we build all three mechanisms on one cluttered 3D benchmark and calibrate against a trivial floor and four classical planners on identical held-out problems. The frame removes five of the six degrees of freedom of SE(3) exactly, at initialisation and at no cost, and is worth +36.5 points. The three mechanisms for the sixth are worth under a point each. Two models that are equally equivariant differ by 28 points. We also bound our own result from above: giving each waypoint local geometry is worth more than the frame is, and the frame still adds +24.8 alongside it.

Lorem	Ipsum	A	B	Δ
dolor	sit amet	00.0	00.0	+00.0
	consectetur	00.0	00.0	+00.0
elit	sed do	00.0	00.0	+00.0
	eiusmod	00.0	00.0	+00.0
tempor	incididunt	00.0	00.0	+00.0
	ut labore	00.0	00.0	+00.0

Table 4: Lorem ipsum dolor sit amet consectetur adipiscing elit

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5.5 The residual, and what it bounds

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Corollary 1 (Lorem ipsum)**.** *Lorem ipsum dolor sit amet consectetur adipiscing elit, sed do eiusmod tempor incididunt* $\|f - \mathcal{A}f\|^2 \le r^2$.

6 Limitations

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7 Conclusion

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The software implementation was carried out with Claude Code under the first author's direction. The implementations of the classical planners used to generate expert trajectories follow the published algorithms [14, 30, 24] and their public reference implementations. The problem formulation, the theoretical analysis, the experimental design and the text of this paper are the authors' own. Every reported number is produced by the released code, whose geometric and equivariance-critical components are verified by property-based tests that run on untrained networks and include negative controls.

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